Math 54 — Week 1

Solutions

P1.

Idea. Clear the first column with row 1, then the second column with row 2; back-substitute.

$$\left[\begin{array}{ccc|c} 1 & 1 & 2 & 9 \\ 2 & 4 & -3 & 1 \\ 3 & 6 & -5 & 0 \end{array}\right] \xrightarrow[R_3-3R_1]{R_2-2R_1} \left[\begin{array}{ccc|c} 1 & 1 & 2 & 9 \\ 0 & 2 & -7 & -17 \\ 0 & 3 & -11 & -27 \end{array}\right] \xrightarrow{2R_3-3R_2} \left[\begin{array}{ccc|c} 1 & 1 & 2 & 9 \\ 0 & 2 & -7 & -17 \\ 0 & 0 & -1 & -3 \end{array}\right]$$

1. The step $2R_3 - 3R_2$ is two row operations: scale row 3 by 2, then add -3 times row 2. Scaling first avoids fractions.
2. The last row says $-x_3 = -3$, so $x_3 = 3$. The second row gives $2x_2 = -17 + 7x_3 = 4$, so $x_2 = 2$. The first row gives $x_1 = 9 - x_2 - 2x_3 = 1$.
3. Check: $1 + 2 + 6 = 9$; $2 + 8 - 9 = 1$; $3 + 12 - 15 = 0$. The solution is $(1, 2, 3)$.

P2.

Idea. A system is inconsistent exactly when an echelon form has a row $[\,0\ 0\ b\,]$ with $b \neq 0$.

1. $R_2 + 2R_1$ gives
$$\left[\begin{array}{cc|c} 1 & -3 & h \\ 0 & 0 & 2h-5 \end{array}\right].$$
2. The second row is the equation $0 = 2h - 5$. The system is consistent exactly when $2h - 5 = 0$, that is, $h = \frac{5}{2}$.

P3.

Idea. After one replacement the second pivot is $9 - 3h$, which may be zero; split into cases.

1. $R_2 - 3R_1$ gives
$$\left[\begin{array}{cc|c} 1 & h & 1 \\ 0 & 9-3h & k-3 \end{array}\right].$$
2. If $h \neq 3$, then $9 - 3h \neq 0$ is a pivot, so both variables are basic and the last column is not a pivot column. Exactly one solution, for every k.
3. If $h = 3$, the second row is $[\,0\ \ 0\ \ k-3\,]$. If $k \neq 3$ it is the equation $0 = k - 3$ with $k - 3 \neq 0$: no solution. If $k = 3$ it is a zero row, and x_2 is free: infinitely many solutions.
4. (a) $h = 3$, $k \neq 3$. (b) $h \neq 3$, any k. (c) $h = 3$, $k = 3$.

P4.

Idea. Forward phase to echelon form, then the backward phase from the rightmost pivot.

1. Forward phase: $R_3 - 2R_1$ gives the first matrix below, and $R_3 - 2R_2$ then gives the second.

$$\left[\begin{array}{ccccc|c} 1 & -2 & -1 & 3 & 0 & -2 \\ 0 & 0 & 1 & -4 & 2 & 5 \\ 0 & 0 & 2 & -8 & 2 & 5 \end{array}\right] \xrightarrow{R_3-2R_2} \left[\begin{array}{ccccc|c} 1 & -2 & -1 & 3 & 0 & -2 \\ 0 & 0 & 1 & -4 & 2 & 5 \\ 0 & 0 & 0 & 0 & -2 & -5 \end{array}\right].$$

   This is an echelon form with pivots in columns 1, 3, 5. The last column is not a pivot column, so the system is consistent.

2. Backward phase, starting with the pivot in column 5: scale row 3 by $-\frac{1}{2}$, then $R_2 - 2R_3$; then clear above the pivot in column 3 with $R_1 + R_2$:

$$\left[\begin{array}{ccccc|c} 1 & -2 & 0 & -1 & 0 & -2 \\ 0 & 0 & 1 & -4 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & \frac{5}{2} \end{array}\right].$$

3. Basic variables x_1, x_3, x_5; free variables x_2, x_4. Solving each row for its basic variable gives the general solution

$$x_1 = -2 + 2x_2 + x_4, \qquad x_2 \text{ free}, \qquad x_3 = 4x_4, \qquad x_4 \text{ free}, \qquad x_5 = \tfrac{5}{2}.$$

4. Check with $x_2 = x_4 = 0$: the point $(-2, 0, 0, 0, \frac{5}{2})$ gives $-2 = -2$, $0 + 5 = 5$, $-4 + 5 = 1$ in the original equations.

P5.

Idea. Use the existence and uniqueness theorem: look for a pivot in the last column, then count free variables.

1. (a) False. Each pivot lies in a different row, so the 3×4 coefficient matrix has at most 3 pivots among its 4 columns, and at least one variable is free. If the system is consistent, it has infinitely many solutions; if it is inconsistent, it has none. In neither case exactly one.

2. (b) False. A counterexample is the system $x_1 = 1$, $x_2 = 2$, $0 = 0$, whose augmented matrix

$$\left[\begin{array}{cc|c} 1 & 0 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{array}\right]$$

   has a zero row and exactly one solution, because neither variable is free.

3. (c) True. Four pivots in a 4×4 coefficient matrix use all four rows, so every row of the augmented matrix has its leading entry in a variable column, and the last column is not a pivot column: the system is consistent for every right-hand side. There are no free variables, so the solution is unique.

P6.

Idea. Each point gives one linear equation in the unknowns a_0, a_1, a_2.

1. The conditions $p(1) = 4$, $p(2) = 9$, $p(3) = 18$ are the equations

$$a_0 + a_1 + a_2 = 4, \qquad a_0 + 2a_1 + 4a_2 = 9, \qquad a_0 + 3a_1 + 9a_2 = 18.$$

2. Row reduce:

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 4 \\ 1 & 2 & 4 & 9 \\ 1 & 3 & 9 & 18 \end{array}\right] \xrightarrow[R_3-R_1]{R_2-R_1} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 4 \\ 0 & 1 & 3 & 5 \\ 0 & 2 & 8 & 14 \end{array}\right] \xrightarrow{R_3-2R_2} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 4 \\ 0 & 1 & 3 & 5 \\ 0 & 0 & 2 & 4 \end{array}\right]$$

3. Back-substitute: $a_2 = 2$, then $a_1 = 5 - 3a_2 = -1$, then $a_0 = 4 - a_1 - a_2 = 3$. So $p(t) = 3 - t + 2t^2$.

4. Check: $p(1) = 3 - 1 + 2 = 4$, $p(2) = 3 - 2 + 8 = 9$, $p(3) = 3 - 3 + 18 = 18$.

P7.

Idea. Four equations in two unknowns may or may not be consistent; row reduce and look for a row $[\,0\ 0\ b\,]$ with $b \neq 0$.

1. Clear the first column:
$$\left[\begin{array}{cc|c} 1 & 2 & 5 \\ 3 & -1 & 1 \\ 2 & 3 & 8 \\ 1 & 1 & 3 \end{array}\right] \xrightarrow[R_4 - R_1]{R_2 - 3R_1,\ R_3 - 2R_1} \left[\begin{array}{cc|c} 1 & 2 & 5 \\ 0 & -7 & -14 \\ 0 & -1 & -2 \\ 0 & -1 & -2 \end{array}\right]$$

2. Scale row 2 by $-\frac{1}{7}$; this gives the first matrix below. Then $R_3 + R_2$ and $R_4 + R_2$ clear the second column below the pivot:
$$\left[\begin{array}{cc|c} 1 & 2 & 5 \\ 0 & 1 & 2 \\ 0 & -1 & -2 \\ 0 & -1 & -2 \end{array}\right] \xrightarrow[R_4 + R_2]{R_3 + R_2} \left[\begin{array}{cc|c} 1 & 2 & 5 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{array}\right]$$

3. There is no row $[\,0\ 0\ b\,]$ with $b \neq 0$, so the system is consistent; both variables are basic, so the solution is unique. Back-substitute: $y = 2$, then $x = 5 - 2y = 1$.

4. Check all four equations: $1 + 4 = 5$; $3 - 2 = 1$; $2 + 6 = 8$; $1 + 2 = 3$.